Problem Set 2
Intermediate Microeconomics
Fall 2022

1. (Short Answers)

   a. Consider a group of people A, B, C and the relation "at least as tall as," as in "A is at least as tall as B." Is this relation transitive? Is it complete?

   b. If the price of good 1 doubles and the price of good 2 triples, does the budget line become flatter or steeper?

   c. Can you explain why taking a positive monotonic transformation of a utility function doesn't change the marginal rate of substitution?

2. (Chapter 3's Exercises 4.4) Mark consumes only cookies and books. At his current consumption bundle, his marginal utility from books is 10 and from cookies is 5. Each book costs $10, and each cookie costs $2. Is he maximizing his utility? Explain. If he is not, how can he increase his utility while keeping his total expenditure constant?

3. (Chapter 3's Exercises 4.14) David's utility function is $U=q_1+2q_2$. Describe his optimal bundle in terms of the prices of q_1 and q_2.

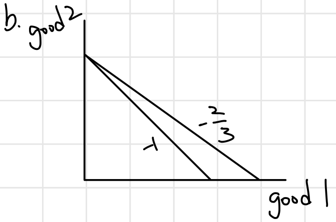
4. "Ernie and Bert, Inc." sells cookies (good x) and bananas (good y). The firm is offering the following deal to its customers. Although the regular prices of cookies and bananas are both fixed at $1 per unit, the first cookie and the first banana that a consumer buys are free. Cookie Monster's utility function is $u(x, y)=x(y+3)$.

   a. Draw Cookie Monster's budget line.
   b. Find his optimal consumption bundle.
   c. What would happen if the regular banana price p_y rises to $2, *ceteris paribus*?

1. a. transitive ⇒ $A \succsim B$, $B \succsim C$ ⇒ $A \succsim C$

complete ⇒ 兩兩皆可互相比較

b.



P_x P_y

2 good1 = 3 good2

$2P_x = 3P_y$ $\quad -\frac{2P_1}{3P_2}$

3 2

$3x + 2y = M$

2good1 = 3good2

relative price ⇒ $-\frac{2P_1}{3P_2} < -1$

same M

c. $u(x,y)$ let $v(u(x,y))$ be a postive monotonic transmation → doesn't chang MRS

$$MRS^u_{xy} = -\frac{\frac{\partial u}{\partial x}}{\frac{\partial u}{\partial y}} \qquad MRS^v_{xy} = -\frac{\frac{\partial v}{\partial u} \times \frac{\partial u}{\partial x}}{\frac{\partial v}{\partial u} \times \frac{\partial u}{\partial y}} = -\frac{\frac{\partial u}{\partial x}}{\frac{\partial u}{\partial y}}$$

2. $MU_{book} = 10$ $\quad P_{book} = 10 = X$ $\quad \frac{MU}{P} = \lambda$ $\quad \frac{MU_{book}}{P_{book}} = \lambda_{book}$ ⇒ $\frac{10}{10} = 1$

$MU_{cookies} = 5$ $\quad P_{cookies} = 2 = Y$

$MRS_{bc} = -\frac{10}{5} = -2$ ⇒ 買到 -5 時停止

$-\frac{P_b}{P_c} = -\frac{10}{2} = -5$ $\quad -2 > -5$

$1 < 2.5$

$\frac{MU_{cookies}}{P_{cookies}} = \lambda_{cookies}$ ⇒ $\frac{5}{2} = 2.5$

$\frac{MU_b}{P_b} = \frac{MU_c}{P_c} = \lambda$

$\frac{MU_b}{MU_c} = \frac{P_b}{P_c}$

⇒ 多買 cookies 少買 books 並同時維持相同的預算

$MU_c - \frac{1}{5}MU_b = \frac{1}{2}MU_b - \frac{1}{5}MU_b > 0$ → 買餅乾的效用比書大，當 $=0$ 時，則兩者獲得效用相同

when $MU_c - \frac{1}{5}MU_b = 0$ ⇒ $MU_c = \frac{1}{5}MU_b$

$5 = \frac{MU_b}{MU_c}$

$-5 = -\frac{MU_b}{MU_c} = MRS_{bc}$

3. $U = q_1 + 2q_2$

s.t. $P_1 q_1 + P_2 q_2 = Y$

$MRS = -\frac{MU_1}{MU_2} = -\frac{P_1}{P_2} \Rightarrow -\frac{1}{2} = +\frac{P_1}{P_2}$

$2P_1 = P_2$

$\Rightarrow P_1 q_1 + 2P_1 q_2 = Y$

$q_1 = \frac{Y - 2P_1 q_2}{P_1}$, $q_2 = \frac{Y - P_1 q_1}{2P_1}$

$MRS = -\frac{1}{2}$ $|MRS| = \frac{1}{2}$

$MRT = -\frac{P_1}{P_2}$ $|MRT| = \frac{P_1}{P_2}$ $\frac{1}{4}$ 4

if $|MRT| < |MRS|$ then $q_1 = \frac{Y}{P_1}$ $q_2 = 0$

if $|MRT| = |MRS|$ then $q_1 \in [0, \frac{Y}{P_1}]$

if $|MRT| > |MRS|$ then $q_1 = 0$ $q_2 = \frac{Y}{P_2}$

4.

4. "Ernie and Bert, Inc." sells cookies (good x) and bananas (good y). The firm is offering the following deal to its customers. Although the regular prices of cookies and bananas are both fixed at \$1 per unit, the first cookie and the first banana that a consumer buys are free. Cookie Monster's utility function is $u(x, y)=x(y+3)$. $Y = 5$ $1 \times (x-1) + 1 \times (y-1) = 5$

a. Draw Cookie Monster's budget line.
b. Find his optimal consumption bundle.
c. What would happen if the regular banana price p_y rises to \$2, *ceteris paribus*?

$xy + 3x$

a. $1 \times (x-1) + 1 \times (y-1) = 5 \Rightarrow x + y = 7$ for $1 \le x \le 6$

$y = 6$ for $x \le 1$

$x = 6$ for $y \le 1$

b. if $(x, y) = (1, 6)$ then $U = 1 \times (6+3) = 9$

$(x, y) = (6, 1)$ then $U = 6 \times (1+3) = 24$

if budget line $\Rightarrow x + y = 7$ for $1 \le x \le 6$

$MRT = -\frac{P_x}{P_y} = -\frac{1}{1} = -1$

$MRS = -\frac{MU_1}{MU_2} = -\frac{y+3}{x}$

$MRT = MRS \Rightarrow +1 = +\frac{y+3}{x}$

$x = y + 3$

$(y+3) + y = 7$

$2y = 4$

$y = 2$ $x = 5$

$U = 5 \times (2+3) = 25 \Rightarrow$ highest utility for $(x, y) = (5, 2)$

c. $1\times(x-1)+2\times(y-1)=5$

$\Rightarrow x+2y=8$ for $1\le x\le 6$

$y=7/2$ for $x\le 1$

$x=6$ for $y\le 1$

if $(x,y)=(1, 7/2)$ then $U= 1\times(3.5+3)=7.5$

$(x,y)=(6, 1)$ then $U=6\times(1+3)=24$

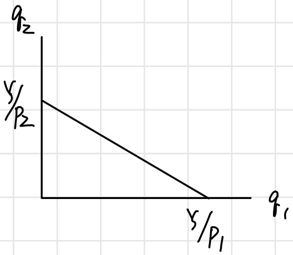
$(x,y)\Rightarrow x+2y=8$ then $MRT=-\frac{1}{2}$

$MRS=-\frac{y+3}{x}$

$MRT=MRT \Rightarrow +\frac{1}{2}=+\frac{y+3}{x} \Rightarrow x=2(y+3)$

$2y+6+2y=8$ $y=1/2$

$x=2\times3.5=7$ $U=7\times(0.5+3)=24.5 \rightarrow$ highest utility b.



$P_1 = 1 \quad \times 2 = 2$

$P_2 = 1 \quad \times 3 = 3$

$Y = 30$

becomes flatter

c. let $u(x,y)$ and $v(u(x,y))$

$$MRS^u_{xy} = -\frac{mu_x}{mu_y}$$

$$MRS^v_{xy} = -\frac{\frac{\partial v}{\partial u} \times \frac{\partial u}{\partial x}}{\frac{\partial v}{\partial u} \times \frac{\partial u}{\partial y}} = -\frac{mu_x}{mu_y}$$

$\Rightarrow$ postive monotonic transformation does not change the MRS

2. 4.4

$mu_b = 10 \quad mu_c = 5$

$$MRS_{bc} = -\frac{10}{5} = -2$$

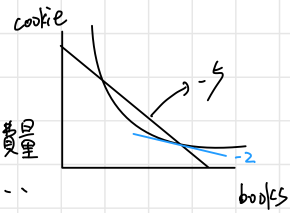
$P_b = 10 \quad P_c = 2$

$$MRT = -\frac{P_b}{P_c} = \frac{-10}{2} = -5$$

$|MRS| < |MRT|$

$\Rightarrow$ 增加 cookie 消費量

減少 book . . .

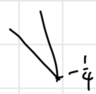


當 $|MRS| = |MRT|$ 時，則可以最大化效用

$MRS_{BC} = -5$

3. 4.14 $U = q_1 + 2q_2$

$-\frac{1}{2} \qquad -\frac{1}{4}$



$$MRS_{1,2} = -\frac{mu_1}{mu_2} = \frac{1}{-2}$$

$$MRT_{1,2} = -\frac{P_1}{P_2}$$

if $|MRS| > |MRT = \frac{P_1}{P_2}|$, then $q_1 = \frac{Y}{P_1}$, $q_2 = 0$

$|MRS| = |MRT = \frac{P_1}{P_2}|$, then $q_1 \in [0, \frac{Y}{P_1}]$

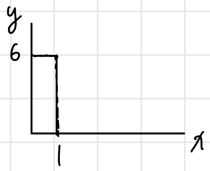
$|MRS| < |MRT = \frac{P_1}{P_2}|$, then $q_1 = 0$, $q_2 = \frac{Y}{P_2}$

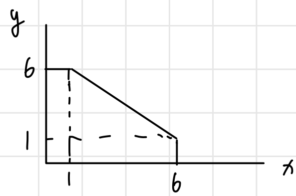
$-\frac{1}{2} \qquad -1$

4. $P_x = 1 \quad P_y = 1 \quad Y = 5$

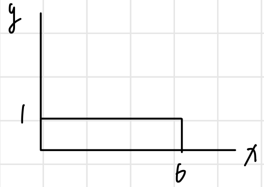
$U = xy + 3x \qquad 1\times(x-1) + (y-1) = 5$

a. if $x \le 1$ then $y = 6$ if $1 \le x \le 6$, the $x+y=7$





if $0 \le y \le 1$, then $x = 6$



b. if $(x,y) = (1,6)$ then $U = 1\times 6 + 3\times 1 = 9$

$(x,y) = (6,1)$ then $U = 6\times 1 + 3\times 6 = 24$

$MRS = -\dfrac{y+3}{x}$ $MRS = MRT \Rightarrow y+3 = x$

$MRT = -1$ $x + y = 7 \Rightarrow 2y + 3 = 7$

$y = 2 \quad x = 5$ $U = 5\times 2 + 5\times 3 = \underset{\sim}{25}$

maximum utility